

Microeconomic Foundations, I: Choice and Competitive Markets

Instructor's Manual

Chapter 12: Producer and Consumer Surplus

Notes on the Chapter

This chapter does precisely what it says it will do: It uses developments of the previous three chapters to make (as) rigorous (as possible) the usual concepts of producer and consumer surplus. The story for producer surplus is simple: As a single price is varied, the derivative of the individual firm's profit function is its netput in that good, and so turning this around, the integral "under" the firm's supply function over any range of prices of the good is the change in the firm's profit over that range of prices. (One must worry, of course, about prices at which the firm's netput set is not singleton and adapt the Fundamental Theorem of Calculus to account for these, but everything works like a charm, albeit this involves pulling a mathematical rabbit about absolute continuity out of a hat.) A warning is given at the end of the discussion about using empirically derived supply functions (which don't hold all other prices fixed), but otherwise this is easy stuff.

And as for consumer surplus: I take the approach of defining the equivalent and compensating variations, relating them to the expenditure function and Hicksian demand functions, and then making enough assumptions (Marshallian demand is single-valued, and the commodity in question is either normal or inferior) so that the conventionally defined change in consumer surplus is trapped between the two variations. I don't try to derive estimates of how different will be the two variations, but point the reader to Willig (1976) on this topic.

After the heavy mathematical slog of the previous three chapters, this all goes down very nicely and easily. It is the rare student (that I've found) who has in active storage the Fundamental Theorem of Calculus for absolutely continuous functions, but the result is so plausible, that it doesn't raise any problems.

Indeed, while I've never had enough time in the quarter to teach Chapter 13 (on aggregation), I think it would be relatively easy in a session on this chapter to use this discussion as a lead-in to the next chapter and perhaps even to get out of the way aggregation of profit-maximizing firms. (The other topics of next chapter probably should be given at least a full class session on their own.)

Notes on the Problems

I'm not particularly enamored of any of the problems at the end of this chapter. I have, in some years, reinforced the warning about empirical versus notional supply functions with a concrete example, taken from an intermediate-level textbook.

Solutions to Non-Starred Problems

■ 12.2. (a) I'll begin by proving a more general result: Suppose $h \in H(p)$ and $h' \in H(p')$, where $H(p)$ is Hicksian demand for some price p and a fixed target utility v . Then $(p - p') \cdot (h - h') \leq 0$.

Proof. Since both h and h' are solutions to the EMP for the target utility level, both are feasible for any prices at that level. Hence, $p \cdot h \leq p \cdot h'$ and $p' \cdot h' \leq p' \cdot h$. Adding these two inequalities gives $p \cdot h + p' \cdot h' \leq p \cdot h' + p' \cdot h$, and you can rearrange terms in the last inequality to get the desired result. ■

To finish part a, note that we are specializing to a case where p' and p'' have identical components except in coordinate position 1, so the inner product $(p' - p'') \cdot (h' - h'')$ is just $(p'_1 - p''_1)(h'_1 - h''_1)$. By assumption $p'_1 > p''_1$, so for the inequality to hold, $h'_1 \leq h''_1$.

(b) We know that $H_1(p_1)$ is nonempty for every value of p_1 . Let $P^* = \{p_1 \in R_{++} : H_1(p_1) \text{ is not singleton}\}$. Since for each $p_1 \in P^*$, $H_1(p_1)$ must contain at least two elements, let $\hat{h}_1(p_1)$ and $\check{h}_1(p_1)$ be any two (distinct) members with $\hat{h}_1(p_1) > \check{h}_1(p_1)$. Then select some rational number $q(p_1)$ from the open interval $(\check{h}_1(p_1), \hat{h}_1(p_1))$. Since $\hat{h}_1(p_1) > \check{h}_1(p_1)$, such a rational can be found.

Now if $p_1, p'_1 \in P^*$ are distinct (that is, $p_1 \neq p'_1$), it must be that $q(p_1) \neq q(p'_1)$. To see this, assume (wlog) that $p_1 > p'_1$. Then $q(p_1) < \hat{h}_1(p_1) \leq \check{h}_1(p'_1) < q(p'_1)$, where the crucial inequality is the one in the middle, which follows from part a.

But this means that q maps points in P^* into the rational numbers in one-to-one fashion, so the cardinality of P^* cannot be larger than the cardinality of the rationals, and there are only countably many rationals.

(c) Part c is so close to identical to the proof of Proposition 12.3 (with inequalities reversed at the end because this is expenditure *minimization* instead of profit *maximization*), that I won't bother to write it out.

(d) Since e is concave, it is absolutely continuous, and so the Fundamental Theorem of Calculus for absolutely continuous functions (as cited for Corollary 12.4) applies.

■ 12.3. If some of the buyers of the good in question are profit-maximizing firms, then their contribution to the overall change in "buyer surplus" will be the change in their profit from the change in the price of this input; the argument used in section 1 for producer surplus applies. While if some sellers of the good are consumers, we would use the equivalent and compensating variations and the same sort of analysis as in Section 12.2 (making assumptions that their supply is single valued and the good for

them is either normal or inferior) to show that their contribution to “change in seller surplus” is bounded on either side by the two variation measures of changes in their welfare. In other words, the analysis of the chapter goes through without a hitch. (No reinterpretation of the profit function is needed, but one does need to broaden the interpretation of the expenditure function and even the “consumption set” of the consumer: If the consumer’s initial endowment of commodities is $z \in R_+^k$, it is most direct to speak of the consumer’s *net trade* x as a point in R^k satisfying $x \geq -z$, where the consumer’s consumption is $x + z$.)

■ 12.5. We are (presumably) fixing all prices except one—which I’ll assume is the first in the list—at levels $p_2, \dots, p_k$, and also income y , and looking at consumer surplus, EV, and CV if p_1 changes from p_1^o to p_1^n . The most natural way to formalize “no income effects” would seem to be something like “Marshallian demand for the good (good 1) doesn’t change as income changes,” but we are only dealing with a single income level, so this doesn’t seem to have any bite. Alternatively, we could formalize “no income effects” as “Hicksian demand is constant in the target utility v , and then we’d be in (easy) business. Proceed as follows:

Let $v^o = \nu((p_1^o, p_2, \dots, p_k), y)$ and $v^n = \nu((p_1^n, p_2, \dots, p_k), y)$. Suppose that, for every p_1 in the interval between p_1^o and p_1^n , the function $v \rightarrow h((p_1, p_2, \dots, p_k), v)$ is constant in v . And suppose that Marshallian demand at income level y is unique for each price $p = (p_1, p_2, \dots, p_k)$ for p_1 between p_1^o and p_1^n . Then consumer surplus as conventionally measured (for this change in the price of good 1) equals $EV = CV$.

This is easily proved: For each p_1 between p_1^o and p_1^n , we know that $\nu((p_1, p_2, \dots, p_k), y)$ lies between v^o and v^n . Therefore, by assumption,

$$h((p_1, p_2, \dots, p_k), v^o) = h((p_1, p_2, \dots, p_k), \nu((p_1, p_2, \dots, p_k), y)) = h((p_1, p_2, \dots, p_k), v^n).$$

But

$$h((p_1, p_2, \dots, p_k), \nu((p_1, p_2, \dots, p_k), y)) = d((p_1, \dots, p_k), y),$$

so we have

$$h((p_1, p_2, \dots, p_k), v^o) = d((p_1, \dots, p_k), y) = h((p_1, p_2, \dots, p_k), v^n).$$

Therefore, the integral of each of these over the range p_1 from p_1^o to p_1^n is the same, which is the desired result. (Note here that by assuming Marshallian demand is unique over this range of arguments, and by assuming that Hicksian demand is constant in v , we are assuming that Hicksian demand is unique over this same range. (I should insert here that we maintain throughout the assumption that the underlying utility function is continuous and locally insatiable, so any Hicksian demand vector is Marshallian demand and vice versa, for corresponding arguments.)

But I'm not entirely happy formalizing "no income effects" in terms of Hicksian demand; the natural meaning of the term, it seems to me, is that Marshallian demand for good 1 is constant in y for each price vector. We can make this work, however. We are fixing p_2 through p_k and varying p_1 between p_1^o and p_1^n . At our originally given y , this gives us $v^o = \nu((p_1^o, p_2, \dots, p_k), y)$ and $v^n = \nu((p_1^n, p_2, \dots, p_k), y)$. Let $\hat{y} = e((p_1^o, p_2, \dots, p_k), v^n)$ and $\check{y} = e((p_1^n, p_2, \dots, p_k), v^o)$. For the sake of concreteness, let me assume that $p_1^o < p_1^n$; we are studying a rise in the price of good 1, so that $v^o > v^n$. Then (in this case) $\hat{y}$ is the income needed to achieve a relatively lower level of utility at a low price for good 1, while $\check{y}$ is the income needed to achieve a relatively high level of utility at high prices. Clearly, in this case, $\hat{y} \leq y \leq \check{y}$. More to the point, suppose we assume that Marshallian demand is unique for all income levels cross prices where income lies between $\hat{y}$ and $\check{y}$ and p_1 is between p_1^o and p_1^n , and that

For all values of p_1 between p_1^o and p_1^n , $y \rightarrow d((p_1, p_2, \dots, p_k), y)$ is constant for y between $\hat{y}$ and $\check{y}$.

Since $h((p_1, p_2, \dots, p_k), v)$ for v between v^o and v^n will equal $d((p_1, p_2, \dots, p_k), y)$ for some y between $\hat{y}$ and $\check{y}$, the italicized assumption just made will ensure that the earlier italicized assumption holds. The key here is that we've taken a range of incomes (from $\hat{y}$ to $\check{y}$) large enough so that, if we vary the target utility level in Hicksian demand between v^o and v^n (fixing all the prices), each value of Hicksian demand corresponds to Marshallian demand for an income level within the range.